Alex Morehead

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Machine Learning • Deep Learning • Computational Biology • High-Performance Computing

Education

University of Missouri — Columbia, Missouri
Ph.D. Computer Science — Aug 2020 - Present

- *Dissertation Advisor*: Prof. Jianlin Cheng
- *Cumulative GPA*: 4.0/4.0
- *Relevant Coursework*: Computational Intelligence, Machine Learning/Pattern Recognition, Neural Networks, Computer Vision, NLP, Unsupervised Learning, Computational Systems Biology, Design & Analysis of Algorithms

Missouri Western State University — St. Joseph, Missouri
B.S. Computer Science — Aug 2016 - May 2020

- *Minor*: Mathematics
- *Cumulative GPA*: 4.0/4.0 with General Studies Honors

Professional Experience

Bioinformatics & Machine Learning Lab — Columbia, Missouri
Graduate Research Assistant | Advisor: Jianlin Cheng — Aug 2020 – Present

- Research and develop novel geometric deep learning algorithms applicable to computational biology
- Authored and published at ICLR 2022 the Geometric Transformer architecture within DeepInteract, a state-of-the-art geometric deep learning pipeline for predicting protein interface contacts
- Released the Equivariant Graph Refinement model for 3D protein complex structural refinement and assessment

Absci AI Research Lab — New York City, New York (Virtual)
AI Scientist Intern | Manager: Joshua Meier — Jun 2022 – Aug 2022

- Researched and built deep learning algorithms for antibody design

NSF Research Experience for Undergraduates in Data Science of Risk and Human Activity — Indianapolis, Indiana
Undergraduate Research Assistant | Program Directors: George Mohler, Mohammad Al Hasan — Jun 2019 - Aug 2019

- Authored an ensemble pipeline of 3 convolutional neural networks trained to detect gunshot sounds in the vicinity
- Deployed the pipeline to a cluster of Raspberry Pi 3 Model B+ microcomputers and evaluated its performance in real-world settings in collaboration with Indianapolis public safety officials
- Published as a manuscript and orally presented the gunshot sound detection project's results at the 2019 IEEE International Conference on Big Data

NSF Research Experience for Undergraduates in Synthetic Biology — St. Joseph, Missouri
Undergraduate Research Assistant | Program Directors: Todd Eckdahl, Jeffrey Poet — May 2018 - Aug 2018

- Investigated applications of combinatorial optimization to model lab experiments performed by synthetic biologists
- Produced Variant Sampler, a Java application for modeling the sample space of in vitro experiments
- Published results of the Variant Sampler project in the American Journal of Undergraduate Research

Center for Academic Support — St. Joseph, Missouri
Computer Science Content Tutor | Manager: Karen Luke — Feb 2017 - Aug 2018

- Tutored a total of 5 undergraduate students, 2 being from underrepresented groups
- Spent up to 1 hour with each student per week, reviewing data structures and object-oriented programming

Publications

• ***Accepted/Published*** •

A Region-Based Deep Learning Approach to Automated Retail Checkout — CVPR '22 – AI City Workshop
M. Shoman, A. Aboah, **A. Morehead**, Y. Duan, A. Daud, Y. Adu-Gyamfi — New Orleans, LA

Geometric Transformers for Protein Interface Contact Prediction — ICLR '22
A. Morehead, C. Chen, J. Cheng — (Virtual)

High-Performance Deep Learning Toolbox for Genome-Scale Prediction of Protein Structure and Function — IEEE SuperComputing '21, St. Louis, MO

M. Gao, P. L. Andersen, **A. Morehead**, S. Mahmud, C. Chen, X. Chen, N. Giri, R. Roy, F. Quadir, T. C. Effler, R. Prout, S. Abraham, W. Elwasif, J. Skolnick, J. Cheng, A. Sedova

Synthetic Biology Bicistronic Designs Support Gene Expression Equally Well in vitro and in vivo — AJUR '20, Journal

O. Koucky, J. Wagner, S. Aguilera, B. Bashaw, Q. Chen, A. Eckdahl, E. Edman, P. Gomez, N. Hanlan, N. Kempf, D. Mattoon, S. McKlin, C. Mazariegos, **A. Morehead**, S. Q. Ong, A. Peterson, M. Rojas, K. Roland, K. Schildknecht, H. Seligmann, K. Slater, A. Tauchen, R. Tittor, T. Travieso, D. Urban, C. Willis, J. Zhou, N. L. Snyder, L. J. Heyer, J. L. Poet, T. T. Eckdahl, A. M. Campbell

Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi — IEEE BigData '19, Los Angeles, CA

A. Morehead, L. Ogden, G. Magee, R. Hosler, B. White, G. Mohler

• *Under Review* •

DRLComplex: Reconstruction of protein quaternary structures using deep reinforcement learning — arXiv '22, Under Review

E. Soltanikazemi, R. S. Roy, F. Quadir, N. Giri, **A. Morehead**, J. Cheng

DProQ: A Gated-Graph Transformer for Protein Complex Structure Assessment — arXiv '22, Under Review

X. Chen, **A. Morehead**, J. Liu, J. Cheng

EGR: Equivariant Graph Refinement and Assessment of 3D Protein Complex Structures — arXiv '22, Under Review

A. Morehead, X. Chen, T. Wu, J. Liu, J. Cheng

3D-Equivariant Graph Neural Networks for Protein Model Quality Assessment — bioRxiv '22, Under Review

C. Chen, X. Chen, **A. Morehead**, T. Wu, J. Cheng

• *Preprint* •

Semi-Supervised Graph Learning Meets Dimensionality Reduction — arXiv '22, Preprint Archive

A. Morehead, W. Chantapakul, J. Cheng

DIPS-Plus: The Enhanced Database of Interacting Protein Structures for Interface Prediction — arXiv '21, Preprint Archive

A. Morehead, C. Chen, A. Sedova, J. Cheng

Presentations

A Region-Based Deep Learning Approach to Automated Retail Checkout — CVPR '22 – AI City Workshop, New Orleans, Lousiana

A. Morehead, M. Shoman, A. Aboah

Geometric Transformers for Protein Interface Contact Prediction — ICLR '22, (Virtual)

A. Morehead

Geometric Transformers for Protein Interface Contact Prediction — Shandong University '22, Qingdao, Shandong Province, China (Virtual)

A. Morehead

High-Performance Toolbox for Deep Learning of Protein Structure and Function — IEEE SuperComputing '21, St. Louis, Missouri (Virtual)

A. Morehead, A. Sedova, M. Gao

Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi — IEEE BigData '19, Los Angeles, California

A. Morehead, L. Ogden, G. Magee

Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi — IUPUI Student Summer Poster Symposium, Indianapolis, Indiana

A. Morehead, L. Ogden, G. Magee

Variant Sampling in vitro with a Scheduling Twist — Alpha Chi National Convention '19, Cleveland, Ohio

A. Morehead

Variant Sampling in vitro with a Scheduling Twist — MWSU PORTAL Summer Research Showcase '18, St. Joseph, Missouri

A. Morehead

Predicting Game Genres by Analyzing Code Structure CSCC Central Plains Conference '18
S. Frazier, **A. Morehead**, S. Prine Maryville, Missouri

Predicting Game Genres by Analyzing Code Structure MWSU Multidisciplinary Research Day '18
S. Frazier, **A. Morehead**, S. Prine St. Joseph, Missouri

Leadership Activities

ICLR Volunteer Apr 2022
NeurIPS Reviewer Sep 2021 – Present
Upsilon Pi Epsilon (ΥΠΕ) | University of Missouri Chapter Aug 2020 - Present
Alpha Chi (AX) | Missouri Western State University Chapter Mar 2018 – May 2020
Kappa Mu Epsilon (KME) | Missouri Western State University Chapter Mar 2018 – May 2020

Awards & Grants

CVPR Travel Grant *Jun 2022*
XSEDE Startup Hardware Grant *Dec 2021*
Dean's Engineering Excellence Fellowship *Aug 2020*
James W. and Joan M. O'Neill Graduate Fellowship in Engineering *Aug 2020*
MWSU Outstanding Graduating Computer Science Student Award *May 2020*
MWSU President's Honor Roll *May 2020*
Floyd Tesmer/Strayer University Prize in Computer Science and Engineering *Apr 2019*
Alpha Chi Region IV Scholarship *Apr 2019*
Grand Midwest Asynchronous Programming Contest 3rd Place Prize *Apr 2017*
East Side Lions Club Scholarship *May 2016*

Projects

BioPython (3.1k Stars) | GitHub: https://github.com/biopython/biopython Apr 2022 – Present
- Created functionality to remove large feature generation temporary files after all processing completes
- Collaborated with senior project developers to ensure proposed solution matched existing code standards

Graphein (300+ Stars) | GitHub: https://github.com/a-r-j/graphein Apr 2022 – Present
- Added workflow to catch illusive OS errors when executing external programs
- Worked with senior project developer to ensure workflow matched existing code standards

EGNN-PyTorch (150+ Stars) | GitHub: https://github.com/lucidrains/egnn-pytorch Apr 2021 – Present
- Noticed the omission of an important geometric unit test for this neural network architecture
- Developed and validated the code to test for node-wise permutation equivariance within the network

DeepInteract (30+ Stars) | GitHub: https://github.com/BioinfoMachineLearning/DeepInteract Oct 2021 – Present
- Designed and developed a state-of-the-art geometric deep learning pipeline for predicting protein interface contacts
- Maintain project by addressing bug fixes, feature requests, and future planning

DIPS-Plus (15+ Stars) | GitHub: https://github.com/BioinfoMachineLearning/DIPS-Plus June 2021 – Present
- Curated (to date) the largest molecular dataset for machine learning of protein interface contacts
- Work to ensure ease of use and adoption by responding to feature requests and open issues

Technical Skills

Programming: Python 3 • Java • Angular 2+ • C/C++ • HTML/CSS • SQL • C# • R • MATLAB

Tools/Frameworks: PyTorch • TensorFlow • NumPy • Pandas • Spring Boot • NodeJS • Docker • AWS • Google Cloud

Methodologies: RESTFUL APIs • Version Control {Git} • Agile Development

News & Media Outreach

[1] MU Engineering. (2020, October 27). *Alex Morehead*. University of Missouri College of Engineering. https://engineering.missouri.edu/alex-morehead/.

[2] MU Engineering. (2021, October 18). *New protein prediction tool could accelerate biological discoveries*. University of Missouri College of Engineering. https://engineering.missouri.edu/new-protein-prediction-tool-could-accelerate-biological-discoveries/.

[3] ORNL. (2021, January 10). *Summit Powers Novel Protein Function Prediction Work*. Oak Ridge National Laboratory. https://www.ornl.gov/news/scientists-use-summit-supercomputer-deep-learning-predict-protein-functions-genome-scale/.

[4] HPCwire. (2021, January 13). *Summit Powers Novel Protein Function Prediction Work*. HPCwire. https://www.hpcwire.com/2022/01/13/summit-powers-novel-protein-function-prediction-work/.

[5] MarkTechPost. (2022, January 18). *Researchers Introduce High-Performance Deep Learning Toolbox for Genome-Scale Prediction of Protein Structure and Function*. MarkTechPost Tech News – AI Paper Summary. https://www.marktechpost.com/2022/01/18/researchers-introduce-high-performance-deep-learning-toolbox-for-genome-scale-prediction-of-protein-structure-and-function/.